

ACTIVITY 1 – FINALS

Test Scenarios and Test Cases Development

Course: Software Testing / Software Quality Assurance

Learning Outcomes

After completing this activity, you should be able to:

- 1. Identify functionalities that require testing in your application.
- 2. Create effective Test Scenarios that describe what should be tested.
- 3. Develop detailed Test Cases from identified Test Scenarios.
- 4. Apply software testing principles to improve software quality.

Part 1: Application Information

Complete the information below:

Item	Details
Application Name	JusticeLink
Application Type/Category	Mobile Application / Legal Tech
Main Features/Modules	1. AI Triage Matching 2. Rights Library

Part 2: Test Scenarios

At least six (6) Test Scenarios for the JusticeLink application:

Format: *TS-XX: Verify that [user action/feature] works correctly.*

Scenario ID	Test Scenario
TS-01	Verify that a user can successfully complete an AI Triage assessment and receive a matching result.
TS-02	Verify that the AI Triage accurately matches the user with appropriate legal resources based on their inputs.
TS-03	Verify that the app gracefully handles errors when the AI Triage service is unavailable or offline.
TS-04	Verify that a user can successfully browse and access categories within the Rights Library.
TS-05	Verify that a user can successfully search for specific legal rights using valid keywords in the Rights Library.
TS-06	Verify that the Rights Library content is displayed accurately and properly formatted.

Part 3: Test Cases

Two Test Scenarios selected (TS-01 and TS-05) with one Positive and one Negative Test Case each.

Template A: Positive / Happy Path Test Case

Related Scenario: TS-01 – Verify that a user can successfully complete an AI Triage assessment and receive a matching result.

Field	Details
Test Case ID	TC_TRIAGE_001
Title	Successful AI Triage Matching
Related Scenario	TS-01
Preconditions	1. The JusticeLink application is installed and accessible. 2. The user has navigated to the AI Triage section. 3. The device has an active internet connection.
Test Steps	1. Tap on the "Start Triage" button. 2. Answer the first triage question with a valid response. 3. Proceed to answer all subsequent triage questions truthfully. 4. Tap the "Submit" or "Get Results" button at the end of the assessment.
Test Data	Category: Employment Law Issue: Wrongful Termination Duration: Last 30 days
Expected Result	1. The application processes the user's input without errors. 2. The user is navigated to the Triage Results screen. 3. The screen displays relevant matched legal resources or actions (e.g., matching lawyers or rights). 4. The recommended resources align with the provided Test Data.

Template B: Negative / Error Path Test Case

Related Scenario: TS-01 – Verify that a user can successfully complete an AI Triage assessment and receive a matching result.

Field	Details
Test Case ID	TC_TRIAGE_002
Title	AI Triage Assessment with No Internet Connection
Related Scenario	TS-01
Preconditions	1. The JusticeLink application is installed and accessible. 2. The user has navigated to the AI Triage section. 3. The device's internet connection is disabled (Airplane Mode or Wi-Fi/Data off).
Test Steps	1. Tap on the "Start Triage" button. 2. Attempt to submit an answer to the first triage question. 3. Alternatively, attempt to submit the final assessment.
Test Data	Network Status: Offline
Expected Result	1. The application fails to retrieve the next question or submit the assessment. 2. A clear error message is displayed (e.g., "Network error. Please check your connection and try again"). 3. The application does not crash. 4. The user remains on the current triage screen and can retry once the connection is restored.

Additional Positive Test Case (TS-05)

Related Scenario: TS-05 – Verify that a user can successfully search for specific legal rights using valid keywords in the Rights Library.

Field	Details
Test Case ID	TC_RIGHTS_001
Title	Successful Rights Library Search with Valid Keyword
Related Scenario	TS-05
Preconditions	1. The JusticeLink application is accessible. 2. The user has navigated to the Rights Library screen. 3. The Rights Library contains entries related to "tenant".
Test Steps	1. Tap on the search bar in the Rights Library. 2. Enter a valid legal keyword (e.g., "tenant"). 3. Tap the Search button on the keyboard or submit the search.
Test Data	Search Keyword: tenant
Expected Result	1. A list of relevant rights and articles matching the keyword is displayed. 2. Results include article titles, summaries, and categories. 3. The number of results is greater than zero. 4. Irrelevant articles (unrelated to the keyword) are not shown.

Additional Negative Test Case (TS-05)

Related Scenario: TS-05 – Verify that a user can successfully search for specific legal rights using valid keywords in the Rights Library.

Field	Details
Test Case ID	TC_RIGHTS_002
Title	Search with No Matching Results (Invalid Keyword)
Related Scenario	TS-05
Preconditions	1. The JusticeLink application is accessible. 2. The user has navigated to the Rights Library screen. 3. The Rights Library does not contain any entries matching the test keyword.
Test Steps	1. Tap on the search bar in the Rights Library. 2. Enter a keyword that does not exist in the legal dictionary or library. 3. Tap the Search button on the keyboard or submit the search.
Test Data	Search Keyword: xyzinvalidright999 (Invalid/Edge Data)
Expected Result	1. No search results are displayed. 2. A clear message is shown such as "No results found" or "No articles match your search". 3. The user remains able to clear the search and perform a new one. 4. The application does not crash or display a critical error screen.

Part 4: Reflection

1. What is the difference between a Test Scenario and a Test Case?

A Test Scenario is a general description of what feature or user activity should be tested. For example, one scenario in the LAYA mobile application is verifying that a citizen can use AI Triage to receive an initial legal assessment. It focuses on the overall function without describing every action.

A Test Case is a detailed procedure for testing that scenario. It specifies the preconditions, test data, step-by-step actions, and expected results. For the AI Triage scenario, a test case may instruct the tester to open the feature, describe a legal concern, answer the follow-up questions, submit the information, and verify that the application presents an appropriate assessment and relevant legal sources. Therefore, a scenario describes what to test, while a test case explains how to test it.

2. Why is it important to create Test Cases before performing software testing?

Creating Test Cases before testing establishes a clear and organized testing process. They ensure that important functions, such as browsing the Rights Library and completing an AI Triage assessment, are tested consistently instead of relying on random exploration. They also define the expected behavior of the application before the results are observed, which helps the tester determine objectively whether a feature passed or failed.

Test Cases also cover both normal and error conditions. For example, the Rights Library should be tested with a valid search query as well as an unavailable research service or a query with no matching results. AI Triage should be tested with sufficient legal information and with incomplete, unclear, or unrelated input. This makes defects and missing requirements easier to identify, allows tests to be repeated after changes, and provides documented evidence that the mobile application's main features were evaluated systematically.

3. Which part of the activity was the most challenging and why?

The most challenging part was creating realistic positive and negative Test Cases for the AI Triage feature. Unlike a basic feature with a fixed output, AI Triage is conversational and its response may vary depending on the user's concern, the completeness of the information provided, the follow-up questions, and the availability of external legal research. The expected result therefore had to focus on consistent requirements—such as asking relevant questions, protecting personal information, providing a clear assessment, presenting verifiable legal sources, and avoiding fabricated legal claims—instead of expecting identical wording every time.

Testing the Rights Library also required careful consideration because it combines curated legal guides with externally retrieved cases and laws. The tests needed to verify not only whether results appeared, but also whether they were relevant, clearly identified as research aids, linked to authoritative sources, and handled safely when no results were found or the external service was unavailable. These challenges required balancing functionality, accuracy, privacy, and user experience while keeping each Test Case specific and repeatable.

